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1)

Definition: Cloud Computing is the delivery of computing services such as servers, storage databases, networking and software over the internet on demand.

Four characteristics:

1)- On - demand self - service: users can access computing resources whenever required without human interaction with service provider

2) Broad network access: Cloud service are accessible through the internet using computers, tablets, and smartphone

3) Resource pooling: providers share computing resource among multiple user using multi-tenancy

4) Rapid elasticity - Resource can be scaled up or down according to workload requirement.

2)

= * Mainframe computing: Centralized computers served multiple users through terminals

* personal computing: Affordable personal computers made computing accessible to individuals.

* Client-server computing: clients requested services over network

* Virtualization - multiple virtual machines could run on a single physical server, improving resource utilization

* internet and web evolution: High-speed internet and web applications enabled remote access to computing services.

Ex: AWS, Microsoft Azure.

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3)

= Role of server virtualization

server virtualization divides a physical server into multiple virtual machines using a hypervisor. Each virtual machine can run its own operating system and application.

Virtualization	Cloud Computing
* Technology that Creates virtual resources * uses hypervisors to run multiple virtual machines * improves hardware utilization	Service model that delivers computing resources over a network use virtualization, automation, networking, and management tool * provides on-demand access and scalable resources

Virtualizat is an important enabling technology while Cloud computing combines technology. to deliver services to users.

4) Discuss the role of data centers in delivering cloud servic

A data center is facility containing servers, systems networking equipment, and supporting infrastructure used to delivers cloud services.

Major Roles:

* Computing : Hosts virtual machines and application
* Storage : Stores databases / files and backup
* Networking : Connects servers and provides internet access
* Security : Use access control servers and provid Internet access
* Availability - User redundant power / cooling and networking to reduce service interruption
* Scalability : provides additional computing resource to meet increasing user demand.

5)

Service Model	Description	User Responsibility	Example
Iaas (infrastructure as a service)	provides virtual Machines Storage and networking	Manages the operating system and applications	Amazon EC2

s (platform s a service	provides a platform for developing and deploying	Manages application Code and data	google app engine
SaaS (Software as a Service	provides ready to-use Software over the internet	Uses the Software and manages user level Setting	google workspace
XaaS (Anything as a Service)	Delivers various IT resources and Capability as Services	Depends on the service provids	Salesforce (CRM as a service)

IaaS provides infrastructure; PaaS provid a develop platfor, SaaS provides ready-to-use Software, and XaaS depends the broder concept of delivering

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